

Week 9 Question Sheet: Explainability for LLMs

1. Which explainability approach is used in AlphaFold [1] and Delphi-2M [2]? How do they suit?

[1] <https://doi.org/10.1038/s41586-021-03828-1>

[2] <https://doi.org/10.1038/s41586-025-09529-3>

2. Is explainability needed in your proposed project? If so, what explainability approach would be suitable for your project?

3. Design to implement an initial version of the explainability.

- For example, testing the explainability in Large Language Models using one of the approaches below
 - Attention visualisation, e.g. BERT-viz
<https://github.com/jessevig/bertviz>
 - SHAP (see notebook on ELE), e.g. for transformers
https://shap.readthedocs.io/en/latest/example_notebooks/text_examples/question_answering/Explaining%20a%20Question%20Answering%20Transformers%20Model.html
 - UMAP, e.g. https://github.com/gerstung-lab/Delphi/blob/main/evaluate_delphi.ipynb in [2]

4. Design an evaluation strategy of the explainability.

What **metrics** will be used? Does it need **annotated data**? Does it need **human evaluation**?